



The coefficient of conformism of a correlative prediction (CCCP): Building up reliable nano-QSPRs/QSARs for endpoints of nanoparticles in different experimental conditions encoded via quasi-SMILES

Alla P. Toropova^{*}, Andrey A. Toropov

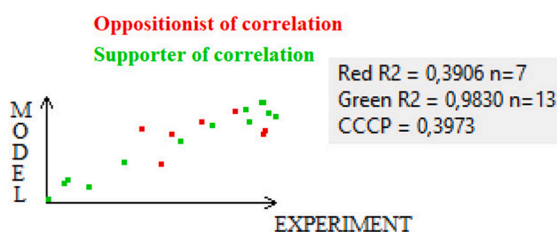
Laboratory of Environmental Chemistry and Toxicology, Istituto di Ricerche Farmacologiche Mario Negri IRCCS, Via Mario Negri 2, 20156 Milano, Italy

HIGHLIGHTS

- Octanol/water partition coefficient and mutagenicity of nanoparticles simulated.
- Quasi-SMILES technique used for the models.
- Coefficient of the conformism of correlative prediction suggested and studied.
- Las Vegas algorithm is considered as a tool to choose better models.

GRAPHICAL ABSTRACT

1. Building of a collection of Monte Carlo models



2. Las Vegas algorithm extracts the Best Monte Carlo algorithm result

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ABSTRACT

Simulation of the physicochemical and biochemical behavior of nanomaterials has its own specifics. However, the main goal of modeling for both traditional substances and nanomaterials is the same. This is an ecologic risk assessment. The universal indicator of toxicity is the n-octanol/water partition coefficient. Mutagenicity indicates the possibility of future undesirable environmental effects, possibly greater than toxicity. Models have been proposed for the octanol/water distribution coefficient of gold nanoparticles and the mutagenicity of silver nanoparticles. Unlike the previous studies, here the models are built using an updated scheme, which includes two improvements. Firstly, the computing involves a new criterion for prediction potential, the so-called coefficient of conformism of a correlative prediction (CCCP); secondly, the Las Vegas algorithm is used to select the potentially most promising models from a group of models obtained by the Monte Carlo algorithm. Apparently, CCCP is a measure of the predictive potential (not only correlation). This can give an advantage in developing a model in comparison to using the classic determination coefficient. Likely, CCCP can be more informative than the classical determination coefficient. The Las Vegas algorithm is able to improve the model obtained by the Monte Carlo method.

^{*} Corresponding author.

E-mail address: alla.toropova@marionegri.it (A.P. Toropova).

1. Introduction

Most phenomena of objective reality can be considered as processes. Any process can be represented as a vector of variables. Variables change their values over time. Sometimes, these changes can be accurately predicted (for example, the behavior of the hands in a working watch). But more often, changes in such variables cannot be predicted accurately (for example, the results of the World Cup). Processes for which all mentioned variables can be precisely determined are deterministic. Processes for which some of the variables cannot be determined precisely are stochastic.

The systematization of knowledge on nanomaterials is a stochastic process. The use of nanomaterials has a variety of consequences, both beneficial (Corsi et al., 2023; Singhal et al., 2022) and alarming (Satish et al., 2022; Scott-Fordsmand and Amorim, 2023). An important, if not key, part of this process is finding ways to simulate the physicochemical and biochemical endpoints associated with nanomaterials in aspect of the environmental risk assessment (García-Quintero and Palencia, 2021; Toropova and Toropov, 2022).

The complexity of developing models for nanomaterials is due to the following circumstances. Firstly, most of the descriptors developed for modeling the properties of conventional (mainly organic substances) cannot be used for nanomaterials. Secondly, in practice, most experimental work with nanomaterials is carried out for a limited number of samples. As a consequence, there are no sufficiently representative databases on nanoparticles, which are necessary to simulation of different endpoint as mathematical functions of the molecular structure. However, the study of endpoints for groups of nanoparticles observed under various experimental conditions can also be considered a task of both theoretical and applied significance (Kleandrova et al., 2014; Luan et al., 2014; Concu et al., 2017; Halder et al., 2020).

In practice, the development of endpoint models for nanoparticles under various experimental conditions can be done using so-called quasi-SMILES. Simplified molecular input-line entry system (SMILES) was applied to develop quantitative structure-property/activity relationships (QSPRs/QSARs) of organic substances. The SMILES is a sequence of symbols which represent the molecular structure. The traditional Monte Carlo method is applied to develop the optimal descriptor of molecules of organic compounds represented by SMILES which is a sum of so-called correlation weights of SMILES-attributes.

The Monte Carlo method algorithm used to obtain optimal descriptors is a random change in correlation weights of SMILES-attributes selected in a random sequence. If a change in the correlation weight improves the target function, it is fixed. Thus, step by step, the target function is improved, ensuring an increase in the predictive potential of the model. The SMILES-attribute can be a chemical element ('C' = carbon; 'O' = oxygen; 'N' = nitrogen etc.), covalent bonds ('=' and '#' are double and triple bonds, respectively), and combinations of elementary attributes (e.g. 'CO', 'NO', 'C=', etc.). Quasi-SMILES is an extension of traditional SMILES by special codes which represent the experimental condition (Toropov and Toropova, 2019b). Indeed, the transition from traditional SMILES to quasi-SMILES greatly expands the scope of optimal random descriptors by allowing the involving of arbitrary eclectic data able to impact the experiment as a basis for model development.

Stochastic processes are a common phenomenon of reality. Stochastic processes are a common phenomenon in practical calculations designed to solve various problems, for instance, related to optimization (Duanmu et al., 2023; Feng et al., 2023; Herrera Velázquez et al., 2023). Stochastic processes can be characterized by varying levels of correlations between the variables involved in changes in some reality or some computation experiments. Correlations can be measured through the magnitude of the correlation coefficient. However, in the practice of the stochastic Monte Carlo calculations designed to model various real quantities, in addition to correlations and probabilities, details often have to be analyzed with appellation to non-correct terms such as "lucky

or unlucky". The Las Vegas algorithm (Tempo and Ishii, 2007) can be used to extract from a group of results observed in a set of attempts to use the Monte Carlo method, the "most lucky" one.

In this study, the Las Vegas algorithm task is remembering the best result (for the calibration set) in the sequence of runs of the Monte Carlo algorithm. The start condition of the procedure of the Monte Carlo optimization is uncertainty (chaos). The final condition of the procedure is 'more or less' good target function value. The more runs of the Monte Carlo algorithm for the Las Vegas algorithm, the better the predictive potential (best target function value) of the final model. Apparently, for the first time, the CORAL software (<http://www.insilico.eu/coral>) is being tested as a tool for implementing the Las Vegas algorithm.

QSPRs/QSARs are a tool to predict or at least to estimate various endpoints related to a variety of substances, i.e. organic, organometallics, polymers, peptides, and nanomaterials. The general scheme of the QSPR/QSAR includes two stages. The first stage is training. The second is verification. The training is the development of a model, a function that, given some available information, calculates the desired endpoint (e.g. octanol/water partition coefficient or mutagenicity). Most currently used models are based solely on one distribution of available data into a training set and a validation set. However, the statistical quality of the prediction of any model likely depends on the distribution of the available data into the above two categories (i) training set; and (ii) validation set (Toropov and Toropova, 2019a). Therefore, to evaluate the model building approach used, it is advisable to test it with several splits of the available data into training and validation sets.

The universal (or popular) indicator of toxicity is the n-octanol/water partition coefficient. Mutagenicity indicates the possibility of future undesirable environmental effects, possibly greater than toxicity. Thus, the above endpoints are quite informative in ecological aspects, since give the possibility to assess the fate of nanoparticles and their ecologic impact. This work studies models of the octanol-water partition coefficient (logP) and mutagenic potential (N_{cp}) for nanoparticles acting under different experimental conditions. These models are built up via optimal descriptors, which are the sums of special correlation weights of the structural features of nanoparticles and experimental conditions. The specified data is input into the program, generating the model using quasi-SMILES. The model's basis is the correlation of optimal descriptors with the simulated endpoint. The values of correlation weights are calculated using the Monte Carlo method. Examples of the use of quasi-SMILES for practical QSPR/QSAR (including Nano-QSPR/QSAR) research are discussed in the book (Edited Toropova and Toropov, 2023b).

In fact, the same approach was used for both endpoints, the important feature of which is the ability to involve experimental conditions in the modeling process. If, subsequently, the list of experimental conditions is expanded, for example, through the use of new equipment, the approach under consideration can be easily adapted to these newly expanded data. Thus the approach is promising from a practical point of view.

It is shown that the predictive potential of these models can be improved if use special parameters control Monte Carlo optimization: the index of the ideality of correlation (*IIC*) (Toropov and Toropova, 2017; Toropova and Toropov, 2017; Kumar et al., 2019; Ahmadi, 2020; Javidfar and Ahmadi, 2020; Goyal et al., 2023) and the correlation intensity index (*CII*) (Toropov and Toropova, 2019b; Ahmadi et al., 2020; Jafari et al., 2020; Kumar and Kumar, 2021; Singh et al., 2023).

In addition to the above *IIC* and *CII*, in this work, a new statistical criterion called the coefficient of conformism of correlative prediction (*CCCP*) is studied. Likely, the *CCCP* is directly related to the predictive potential of a model (not only "classic" correlation). However, the use of the listed indexes gives an improvement of the statistical quality of a model for the calibration set (as well as often for external validation set) in detriment the active- and passive training sets.

Thus, the aim of this study is an attempt to construct models of the physicochemical (logP) and biochemical (mutagenicity) behavior of

nanoparticles through the Monte Carlo method using a new criterion of the predictive potential (CCCP). For improving the models, the Las Vegas algorithm which selects the best results (for the calibration set) in a series of runs of the stochastic (Monte Carlo technique) optimization process is studied. Both, the CCCP and the Las Vegas algorithm as possible improvements for the CORAL software are assessed in the first time.

2. Method

2.1. Data

Two databases related to the logP of gold nanoparticles and the mutagenicity of silver nanoparticles are considered here.

2.1.1. Data on octanol/water partition coefficient (logP)

The experimental data on octanol/water partition coefficient (logP) for gold nanoparticles under different experimental conditions taken from the literature (Yan et al., 2019). Fig. 1 contains an example of the construction of quasi-SMILES used to the simulation of logP (Toropova and Toropov, 2021).

2.1.2. Data on mutagenicity

Analogically, the experimental data on mutagenicity is expressed as the number of colonies per plate (Ncp) taken from the literature (Guo et al., 2016). Quasi-SMILES for these data on the mutagenicity of silver nanoparticles suggested in (Toropov and Toropova, 2019b). However, a more convenient to develop a model version of quasi-SMILES is used here (Table 1).

To develop models, available data on both endpoints were divided into four special sub-sets, namely, active training ($\approx 25\%$), passive training ($\approx 25\%$), calibration ($\approx 25\%$), and validation ($\approx 25\%$) sets. Five random distributions of the data into the above sub-sets were used as the basis to build up models for both logP and mutagenicity.

Table 2 lists the percentage of identity of splits for both endpoints considered. One can see that the splits are far from being identical.

Models for both endpoints were built according to the same scheme, the main result of which can be expressed by the equation

$$\text{Endpoint} = C_0 + C_1 \times DCW(T, N) \quad (1)$$

C_0 and C_1 are the regression coefficients. $DCW(T, N)$ is the optimal descriptor calculated with the correlation weights, which are obtained by the Monte Carlo method, which is aimed to get the largest value of a target function described below. T is the threshold to separate the rare quasi-SMILES codes, which are removed from the simulation process, and active (non-rare) codes, which are the basis for the definition of the optimal descriptor value. N is the number of epochs (iterations) of the Monte Carlo optimization.

2.2. Optimal descriptors

2.2.1. Optimal descriptor for simulation logP of gold nanoparticles

Models for logP for gold nanoparticles built up using the optimal descriptors calculated as

Table 1

The generalized codes of quasi-SMILES, which are used to simulate mutagenicity (N_{cp}) for silver nanoparticles.

Categories	List of experimental conditions	Codes used to construct quasi-SMILES
Mutagenicity	TA100 →	[TA100]
	TA98 →	[TA98]
Coating	20-nm citrate →	[20cit]
	20-nm PVP →	[20PVP]
	50-nm citrate →	[50cit]
	50-nm PVP →	[50PVP]
	100-nm citrate →	[100cit]
	100-nm PVP →	[100PVP]
Dose ($\mu\text{g}/\text{plate}$)	0.0 →	[d0]
	6.3 →	[d6,3]
	12.5 →	[d12,5]
	25 →	[d25]
	50 →	[d50]
	100 →	[d100]

Table 2

The percentage of identity for five splits in the training and validation sets.

	1	2	3	4	5
Octanol/water partition coefficient for gold nanoparticles, logP (Yan et al., 2019)					
1*	100	23.8	33.3	48.8	42.9
2	42.9	100	42.9	24.4	42.9
3	24.4	43.9	100	39.0	47.6
4	33.3	28.6	34.1	100	39.0
5	33.3	33.3	43.9	57.1	100
Mutagenicity is the number of colonies per plate, Ncp (Guo et al., 2016)					
1	100	38.9	33.3	27.8	33.3
2	44.4	100	38.9	22.2	33.3
3	55.6	50.0	100	38.9	50.0
4	27.8	44.4	38.9	100	50.0
5	27.8	38.9	33.3	33.3	100

* If $i > j$, then the matrix element $[i, j]$ means the percentage of identity for the active training sets. If $i < j$, then the matrix element $[i, j]$ means the percentage of identity for the validation sets (external sets). The i and j mean the numbering of 5 splits examined.

$$DCW(3, 15) = \sum CW(S_k) + \sum CW(SS_k) \quad (2)$$

S_k is an atom of quasi-SMILES, i.e. a group of symbols of quasi-SMILES that cannot be considered separately, e.g. 'C', 'N', 'Cl', '=', '[5]', '[310]', etc.; SS_k are compositions of two symbols of usual SMILES involved into the quasi-SMILES, e.g. 'CN', 'O=', etc.

2.2.2. Optimal descriptor for simulation mutagenicity of silver nanoparticles

Models for mutagenic potential (N_{cp}) for silver nanoparticles built up using the optimal descriptors calculated as

$$DCW(1, 15) = \sum CW(S_k) \quad (3)$$

S_k can be one of the codes demonstrated in Table 1.

ID	Size [nm]	Frequency of ligand 1	Frequency of ligand 2	SMILES of ligand 1	SMILES of ligand 2	logP
↓	↓	↓	↓	↓	↓	↓
GNP57	[5,8]	[31]	[101]	S1SC(CCl)CCCC(NCCCN)=O	S1SC(CCl)CCCC(N)=O	2.680

↓

```
#GNP57 [5,8][31][101]S1SC(CCl)CCCC(NCCCN)=O.S1SC(CCl)CCCC(N)=O -2.680
-GNP58 [5,8][82][75]S1SC(CCl)CCCC(NCCCN)=O.S1SC(CCl)CCCC(N)=O -2.350
+GNP59 [5,8][144][0]S1SC(CCl)CCCC(NCCCN)=O.[NoSMS2] -1.740
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Fig. 1. The general scheme of quasi-SMILES used to simulate logP for gold nanoparticles (GNP).

2.3. Monte Carlo algorithm

The Monte Carlo optimization is a tool to calculate the correlation weights. Here, a group of target functions for the Monte Carlo optimization is compared. The first from the group of target functions is the following:

$$TF_0 = r_{AT} + r_{PT} - |r_{AT} - r_{PT}| \times 0.1 \quad (4)$$

r_{AT} and r_{PT} are correlation coefficients between the observed and predicted endpoints for the active and passive training sets, respectively.

In several works, *IIC* was used to improve the predictive potential of models calculated from optimal descriptors. It is calculated using the structured training set that includes (i) the active training sub-set, (ii) the passive training sub-set, and (iii) the calibration set.

IIC is calculated with data on the calibration set as follows (Toropov and Toropova, 2017):

$$IIC = r_c \frac{\min(-MAE_C, +MAE_C)}{\max(-MAE_C, +MAE_C)} \quad (5)$$

$$\min(x, y) = \begin{cases} x, & \text{if } x < y \\ y, & \text{otherwise} \end{cases} \quad (6)$$

$$\max(x, y) = \begin{cases} x, & \text{if } x > y \\ y, & \text{otherwise} \end{cases} \quad (7)$$

$$-MAE_C = \frac{1}{-N} \sum |\Delta_k|, \quad -N \text{ is the number of } \Delta_k < 0 \quad (8)$$

$$+MAE_C = \frac{1}{+N} \sum |\Delta_k|, \quad +N \text{ is the number of } \Delta_k \geq 0 \quad (9)$$

$$\Delta_k = \text{observed}_k - \text{calculated}_k \quad (10)$$

The observed and calculated are corresponding values of the endpoint.

The second considered target function for the Monte Carlo optimization is the following:

$$TF_1 = TF_0 + IIC \times 0.3 \quad (11)$$

Another possibility to improve the predictive potential of a model building up the Monte Carlo technique is to use *CII* (Toropov and Toropova, 2019b). *CII* was calculated using data on the calibration set as follows:

$$CII = 1 - \sum \text{Protest}_k \quad (12)$$

$$\text{Protest}_k = \begin{cases} R_k^2 - R^2, & \text{if } R_k^2 - R^2 > 0 \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

R^2 is the correlation coefficient for a set that contains n substances. R_k^2 is the correlation coefficient for $n-1$ substances of a set after removing of k -th substance. Hence, if the $\Delta = (R_k^2 - R^2)$ is larger than zero, the k -th substance is an “oppositionist” for the correlation between experimental and predicted values of the set. A small sum of “protests” means a more “intensive” correlation.

The third considered target function for the Monte Carlo optimization is the following:

$$TF_2 = TF_0 + CII \times 0.3 \quad (14)$$

However, in addition to the above-mentioned “oppositionists” of the correlation, there are also its “supporters” for whom the $\Delta = (R_k^2 - R^2)$ is less than zero. The comparison of correlation coefficients separately for “supporters” and “oppositionists” of the correlation can give the ratio of these values that may be an informative criterion for the Monte Carlo optimization, similarly to tested up *IIC* and *CII*.

$$CCCP = \frac{\sum \Delta R(\text{oppositionist})}{\sum \Delta R(\text{supporter})} \quad (15)$$

The suggested ratio is able to show the measure of conformism of the “oppositionists” and “supporters”. Thus, one can name this criterion the conformism coefficient of correlative prediction (*CCCP*).

$$TF_3 = TF_0 + CCCP \times 0.3 \quad (16)$$

The levels of correlation predictions may likely be controlled in a stochastic Monte Carlo process, taking into account both the above-mentioned special quantities *IIC*, *CII*, and *CCCP*.

Fig. 2 shows an example of the mutual positions of “oppositionists” and “supporters” observed for a correlative prediction of logP of gold nanoparticles.

Fig. 3 shows that the statistical criteria mentioned are varied. However, a certain similarity can be identified, for instance, between *CCC* and *IIC*. Both of these criteria decrease greatly if the dots in coordinates the experiment versus predicting deviate from the diagonal.

One can also talk about a certain similarity between the values of *CII* and *CCCP* for different levels of dispersion in plots of the experiment versus prediction. In the case of anti-correlations, the numerical values of the *CCCP* in contrast to all other considered criteria increase greatly, as can be seen from the diagram presented in Fig. 4.

The main task of the Monte Carlo algorithm considered here is to provide the correlation weights for calculating an endpoint model via Eq. (1). Fig. 5 represents the general scheme of the Monte Carlo algorithm run.

Fig. 6 contains averaged images of the Monte Carlo optimization histories for modeling the octanol/water distribution coefficient and the mutagenicity. In both cases, pictures related to split #1.

One can see that preferable the statistical quality of these models is observed for target function TF_3 . At the initial stages of the Monte Carlo optimization process (up to approximately the fifth epoch), changes in

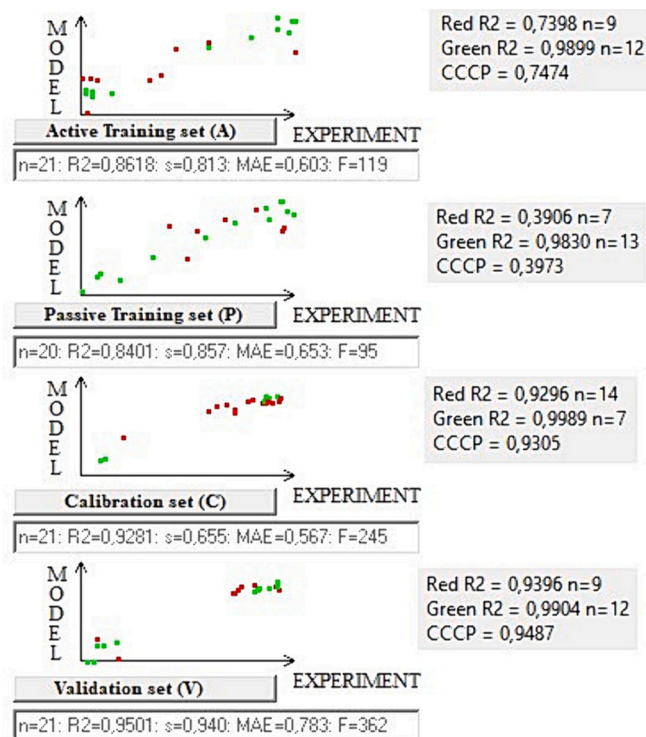


Fig. 2. An example of the location of points that are “oppositionists” for a correlation (shown in red) and points that are “supporters” of a correlation (shown in green) in four sub-sets (active training set, passive training set, calibration set, and validation set) used to simulate logP of gold nanoparticles.

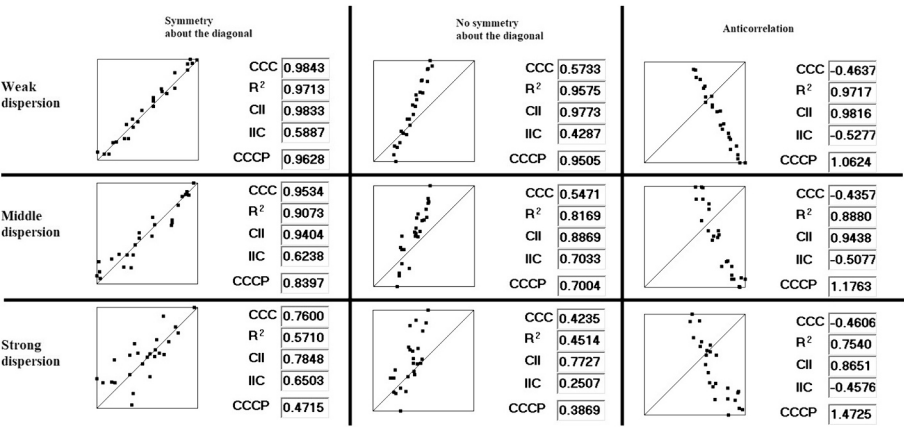


Fig. 3. The comparison of different statistical criteria for various correlations (anti-correlations).

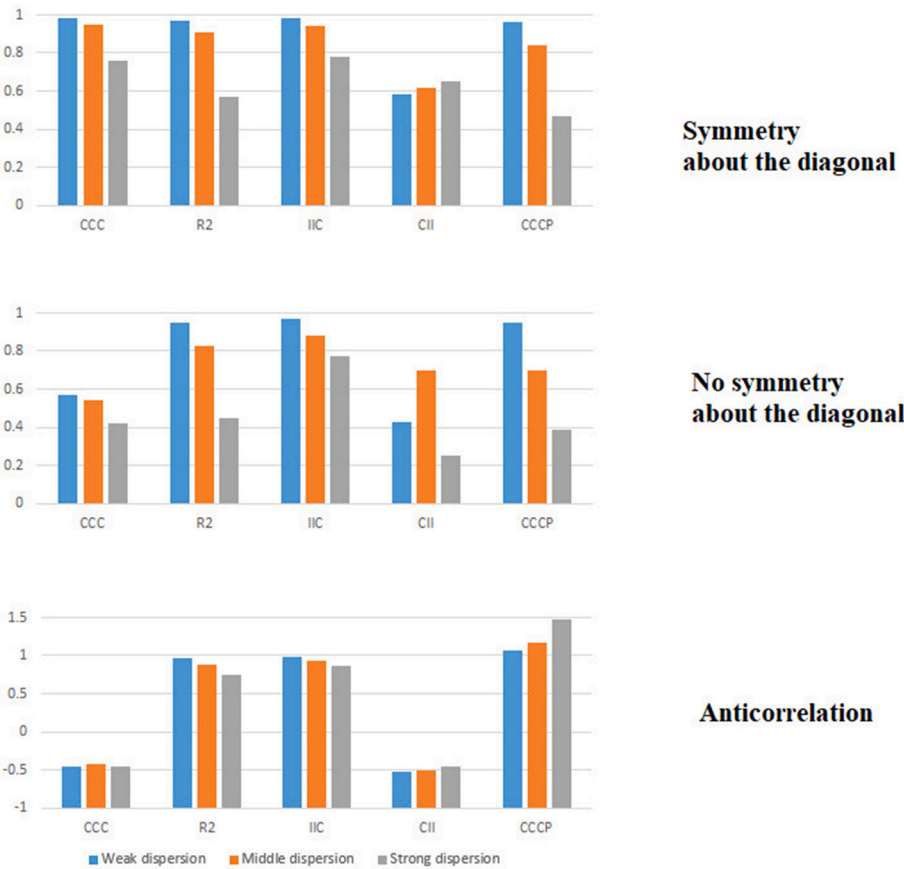


Fig. 4. Analysis of reaction of considered criteria to different kind correlations varied in the level of dispersion (weak, middle, and strong) as well in the case of anti-correlations.

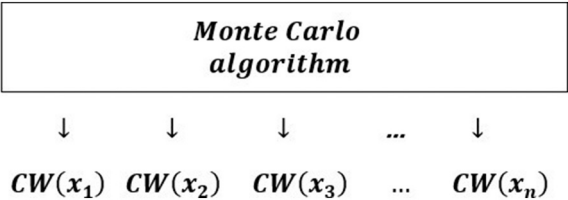


Fig. 5. The general scheme of the Monte Carlo algorithm which provides the correlation weights for a model.

the coefficient of determination over the four sets are chaotic for TF_1 and TF_2 . This is observed for both endpoints considered. In the case of TF_3 , the optimization process for both endpoints is stable almost from the very beginning, but most importantly, in this case, the highest values of the coefficient of determination are observed for the validation sets. It should be noted, however, that for the logP models, there is some overfitting for TF_2 and TF_3 (a decrease in the coefficient of determination for the validation set). However, these overfittings are insignificant and, in general, the final models (after 15 optimization epochs) are quite acceptable.

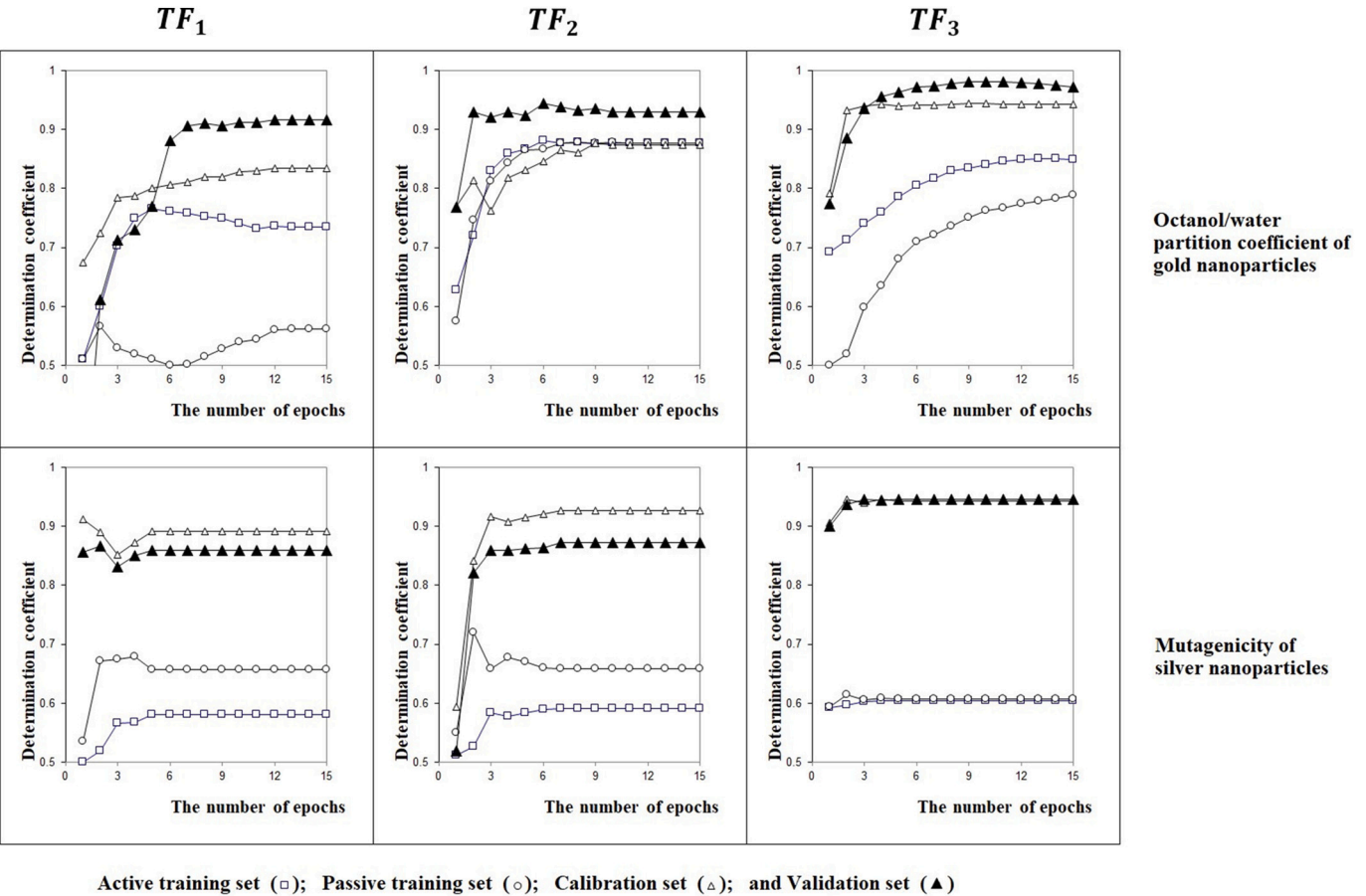


Fig. 6. Histories of the Monte Carlo optimization for different target functions in the simulation of the logP and mutagenicity.

2.4. Las Vegas algorithm

A randomized algorithm that always succeeds in producing a correct output but whose running time depends on random events is known as a Las Vegas algorithm (Tempo and Ishii, 2007). Here, the Las Vegas algorithm is applied to extract models with the best predictive potential from a group of runs of the Monte Carlo method. Fig. 7 represents the general scheme of the Las Vegas algorithm results obtained.

In other words, the Las Vegas algorithm extracts from several Monte Carlo optimizations and runs the result with the best statistics for the calibration set. This extraction is based on the hypothesis that if the coefficient of determination for the calibration set is high, it is very likely that the coefficient of determination for the validation set will also be high.

3. Results and discussion

3.1. Simulation of octanol/water partition coefficient of gold nanoparticles

Table 3 contains statistical characteristics of the octanol/water partition coefficient models for five random splits into training and validation sets. It can be seen that for all models built using different splits, there is an improvement in the predictive potential from models built using the target function TF_1 to models built using the target function TF_3 . The target function TF_2 gives some intermediate results; that is, it shows better statistics than the TF_1 -model, but the statistics are worse than the TF_3 -model. Consequently, using CCCP in the Monte Carlo optimization improves the predictive potential of models for logP.

3.2. Simulation mutagenicity of silver nanoparticles

Table 4 contains the statistical characteristics of models for the

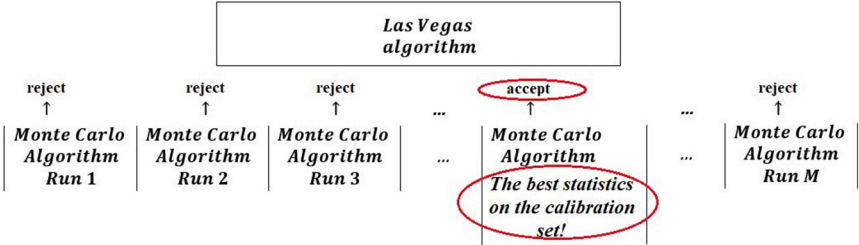


Fig. 7. The general scheme of the Las Vegas algorithm used to build the models.

Table 3

The statistical characteristics of models for logP of gold nanoparticles.

Target function	Split	Set*	<i>n</i>	<i>R</i> ²	<i>CCC</i>	<i>IIC</i>	<i>CII</i>	<i>Q</i> ²	<i>MAE</i>	<i>F</i>
<i>TF</i> ₁	1	A	21	0.5797	0.7339	0.6922	0.7281	0.5072	1.13	26
		P	20	0.5192	0.6868	0.5488	0.6870	0.4321	1.23	19
		C	21	0.8608	0.9163	0.8949	0.9129	0.8277	0.488	117
		V	21	0.8597	–	–	–	–	0.48	–
	2	A	21	0.5733	0.7288	0.5679	0.7635	0.5012	1.04	26
		P	20	0.6008	0.7466	0.6628	0.7202	0.5232	1.01	27
		C	21	0.6916	0.7832	0.8310	0.8322	0.6331	0.753	43
		V	21	0.8851	–	–	–	–	0.72	–
	3	A	21	0.7008	0.8241	0.7611	0.7964	0.6510	0.873	45
		P	21	0.7926	0.8776	0.5593	0.8494	0.7599	0.742	73
		C	21	0.8557	0.9049	0.9235	0.8731	0.8326	0.540	113
		V	20	0.8269	–	–	–	–	0.54	–
	4	A	20	0.8098	0.8949	0.8999	0.8715	0.7746	0.766	77
		P	21	0.5881	0.7338	0.6363	0.7590	0.5261	1.14	27
		C	21	0.5521	0.7134	0.7421	0.6945	0.4556	0.846	23
		V	21	0.8447	–	–	–	–	0.60	–
	5	A	21	0.7095	0.8300	0.5183	0.8031	0.6611	0.949	46
		P	21	0.7190	0.8147	0.3444	0.8387	0.6631	0.947	49
		C	20	0.7288	0.8303	0.8452	0.7951	0.6759	0.622	48
		V	21	0.8830	–	–	–	–	0.69	–
<i>TF</i> ₂	1	A	21	0.9102	0.9530	0.8673	0.9494	0.8935	0.500	193
		P	20	0.9145	0.9520	0.6420	0.9332	0.9002	0.557	193
		C	21	0.8559	0.8802	0.5712	0.9113	0.8294	0.694	113
		V	21	0.8712	–	–	–	–	0.70	–
	2	A	21	0.9168	0.9566	0.8705	0.9448	0.9023	0.478	209
		P	20	0.9095	0.9335	0.4588	0.9333	0.8932	0.521	181
		C	21	0.8306	0.8411	0.4337	0.9147	0.7921	0.679	93
		V	21	0.9376	–	–	–	–	0.80	–
	3	A	21	0.8923	0.9431	0.7085	0.9259	0.8752	0.519	157
		P	21	0.9170	0.9551	0.5455	0.9481	0.9026	0.552	210
		C	21	0.8690	0.8888	0.4435	0.9072	0.8459	0.643	126
		V	20	0.8925	–	–	–	–	0.81	–
	4	A	20	0.9091	0.9524	0.7801	0.9342	0.8918	0.499	180
		P	21	0.9207	0.9504	0.9557	0.9426	0.9072	0.575	220
		C	21	0.8219	0.8738	0.4253	0.8863	0.7905	0.592	88
		V	21	0.8257	–	–	–	–	0.74	–
	5	A	21	0.8819	0.9372	0.8537	0.9235	0.8619	0.604	142
		P	21	0.9154	0.9464	0.7163	0.9497	0.9008	0.579	206
		C	20	0.8337	0.8636	0.2349	0.9296	0.7798	0.589	90
		V	21	0.8665	–	–	–	–	0.63	–
<i>TF</i> ₃	1	A	21	0.8618	0.9258	0.8440	0.9043	0.8362	0.603	119
		P	20	0.8401	0.8979	0.5002	0.8836	0.8161	0.653	95
		C	21	0.9281	0.8919	0.2711	0.9537	0.9083	0.567	245
		V	21	0.9501	–	–	–	–	0.78	–
	2	A	21	0.8816	0.9371	0.7042	0.9306	0.8570	0.562	141
		P	20	0.8071	0.8890	0.5169	0.9046	0.7687	0.807	75
		C	21	0.8759	0.8346	0.1741	0.9409	0.8405	0.828	134
		V	21	0.9423	–	–	–	–	0.93	–
	3	A	21	0.8284	0.9062	0.8274	0.8997	0.8011	0.643	92
		P	21	0.8451	0.8901	0.4565	0.9171	0.8151	0.858	104
		C	21	0.8821	0.8872	0.4835	0.9361	0.8391	0.551	142
		V	20	0.9483	–	–	–	–	0.58	–
	4	A	20	0.8616	0.9256	0.7594	0.9097	0.8363	0.656	112
		P	21	0.8975	0.9299	0.6797	0.9301	0.8757	0.678	166
		C	21	0.8751	0.9221	0.9136	0.9314	0.8427	0.497	133
		V	21	0.9276	–	–	–	–	0.54	–
	5	A	21	0.8318	0.9082	0.6840	0.8823	0.8005	0.695	94
		P	21	0.8080	0.8665	0.5595	0.8771	0.7818	0.851	80
		C	20	0.8953	0.8769	0.1006	0.9623	0.8548	0.576	154
		V	21	0.9297	–	–	–	–	0.91	–

* A = active training set; P = passive training set; C = calibration set; V = validation set; *n* = the number of quasi-SMILES; *R* = correlation coefficient; *CCC* = concordance correlation coefficient; *IIC* = index of ideality of correlation; *CII* = correlation intensity index; *Q* = cross validated *R*; *MAE* = mean absolute error; *F* = Fischer F-ratio.

mutagenicity of silver nanoparticles. In fact, one can see the same trends that were observed for the logP models. Thus, in the case of mutagenicity simulation, the *CCCP* also improves the predictive potential of models.

3.3. Best model for logP

The best predictive potential, that is, the largest value of the coeffi-

cient of determination on the validation set for octanol-water partition coefficient models, is observed in the model built using the first split. This model is as follows:

$$\log P = -4.397 + 0.4007 \cdot DCW(3,15) \quad (16)$$

3.4. Best model for mutagenicity

The best predictive potential, that is, the largest value of the coeffi-

Table 4
The statistical characteristics of models for mutagenicity of silver nanoparticles.

Target function	Split	Set*	n	R ²	CCC	IIC	CII	Q ²	MAE	F
TF ₁	1	A	18	0.5379	0.6995	0.5867	0.8006	0.3300	36.8	19
		P	18	0.6780	0.7042	0.3335	0.8143	0.5571	35.3	34
		C	18	0.8454	0.9147	0.9085	0.9208	0.8036	12.6	88
		V	18	0.7883	–	–	–	–	18.62	–
	2	A	18	0.6238	0.7683	0.6318	0.7689	0.4962	38.2	27
		P	18	0.5860	0.6862	0.4937	0.7946	0.4022	36.5	23
		C	18	0.7767	0.8770	0.8717	0.8494	0.7348	14.8	56
		V	18	0.8219	–	–	–	–	13.94	–
	3	A	18	0.5938	0.7451	0.7706	0.8230	0.4601	39.7	23
		P	18	0.6345	0.7280	0.6870	0.7872	0.4683	36.4	28
		C	18	0.8590	0.8975	0.9213	0.9358	0.8115	14.3	97
		V	18	0.8078	–	–	–	–	14.90	–
	4	A	18	0.6700	0.8024	0.6548	0.7940	0.5436	33.6	32
		P	18	0.7991	0.6252	0.7515	0.8692	0.6971	46.6	64
		C	18	0.8323	0.7939	0.9114	0.8829	0.7944	27.4	79
		V	18	0.8218	–	–	–	–	27.04	–
	5	A	18	0.5164	0.6811	0.4573	0.8054	0.3061	38.2	17
		P	18	0.6403	0.7368	0.6877	0.8333	0.5299	35.8	28
		C	18	0.8651	0.9295	0.9288	0.9224	0.8325	12.6	103
		V	18	0.9207	–	–	–	–	9.03	–
TF ₂	1	A	18	0.5940	0.7453	0.6166	0.8158	0.4265	33.3	23
		P	18	0.6514	0.7164	0.5082	0.7977	0.5160	38.5	30
		C	18	0.9410	0.9455	0.5336	0.9693	0.9234	9.21	255
		V	18	0.9004	–	–	–	–	11.25	–
	2	A	18	0.6513	0.7889	0.6456	0.8086	0.5338	38.9	30
		P	18	0.6474	0.7095	0.8043	0.7957	0.4652	38.4	29
		C	18	0.8550	0.8557	0.2812	0.9125	0.8258	18.6	94
		V	18	0.8352	–	–	–	–	15.57	–
	3	A	18	0.6258	0.7698	0.7911	0.8160	0.4948	37.4	27
		P	18	0.6465	0.7238	0.6497	0.8141	0.4871	36.7	29
		C	18	0.8575	0.8758	0.7479	0.9347	0.8156	16.6	96
		V	18	0.8877	–	–	–	–	12.94	–
	4	A	18	0.6860	0.8137	0.8282	0.8059	0.5576	32.9	35
		P	18	0.8087	0.6296	0.8277	0.8676	0.7090	46.1	68
		C	18	0.8076	0.7808	0.7986	0.8756	0.7626	28.0	67
		V	18	0.8407	–	–	–	–	27.36	–
	5	A	18	0.5715	0.7274	0.6048	0.7684	0.3628	36.1	21
		P	18	0.6647	0.7700	0.5641	0.8287	0.5506	36.8	32
		C	18	0.9236	0.9505	0.8319	0.9620	0.9017	10.6	193
		V	18	0.8126	–	–	–	–	13.11	–
TF ₃	1	A	18	0.6112	0.7587	0.6255	0.8118	0.4508	33.0	25
		P	18	0.6001	0.7120	0.5849	0.7923	0.4490	41.8	24
		C	18	0.9567	0.9552	0.7982	0.9818	0.9426	8.20	354
		V	18	0.9365	–	–	–	–	7.69	–
	2	A	18	0.5987	0.7490	0.6190	0.8217	0.4632	44.0	24
		P	18	0.5971	0.7161	0.5706	0.8214	0.4058	38.9	24
		C	18	0.9371	0.8824	0.9680	0.9647	0.9228	17.1	238
		V	18	0.8932	–	–	–	–	13.88	–
	3	A	18	0.5971	0.7477	0.6182	0.8053	0.4625	39.7	24
		P	18	0.6527	0.7270	0.6318	0.7981	0.4953	34.4	30
		C	18	0.9390	0.9591	0.6644	0.9746	0.9193	8.30	246
		V	18	0.9096	–	–	–	–	8.60	–
	4	A	18	0.6726	0.8043	0.6561	0.8178	0.5581	36.2	33
		P	18	0.6782	0.6358	0.6659	0.8167	0.4944	43.5	34
		C	18	0.8802	0.8608	0.8342	0.9544	0.8430	17.8	118
		V	18	0.9076	–	–	–	–	13.23	–
	5	A	18	0.5695	0.7257	0.7546	0.7712	0.3642	37.0	21
		P	18	0.6321	0.7564	0.5094	0.8115	0.5052	39.8	27
		C	18	0.9554	0.9771	0.9440	0.9757	0.9433	7.01	343
		V	18	0.9518	–	–	–	–	7.98	–

* A = active training set; P = passive training set; C = calibration set; V = validation set; n = the number of quasi-SMILES; R = correlation coefficient; CCC = concordance correlation coefficient; IIC = index of ideality of correlation; CII = correlation intensity index; Q = cross validated R; MAE = mean absolute error; F = Fischer F-ratio.

cient of determination on validation set for mutagenicity (the number of colonies per plate) models, is observed in the model built using the fifth split. This model is as follows:

$$N_{cp} = 5.332 + 26.11 \cdot DCW(I,15) \tag{17}$$

Supplementary materials section contains technical details on the best models for (i) the octanol/water partition coefficient of gold nanoparticles and (ii) the mutagenicity of silver nanoparticles. The general

format of results for both endpoints is the following. The first is the random split of available data into the active- and passive training sets, calibration set, and validation set (Table S1 and Table S3 for logP and mutagenicity, respectively); and the second is the lists of the correlation weights values (Table S2 and Table S4 for logP and mutagenicity, respectively). The applicability domain and the frequencies of quasi-SMILES codes also are represented in the Supplementary materials section.

3.5. Why does CCCP improve predictive potential?

A comparison of the calculation schemes for the three considered predictive potential criteria shows that the CCCP has nothing in common with the IIC.

However, the new criterion has some similarity to CII, built exclusively on objects (quasi-SMILES) that are “oppositionists” to correlation in the above sense. The difference between the CCCP and the IIC is the simultaneous consideration of the influence of both “oppositionists” and “supporters” of correlation. Considering the results of the described computational experiments, it seems namely this difference that makes CCCP a one more reliable indicator of the way to improving predictive potential.

3.6. CORAL software and quasi-SMILES

Quasi-SMILES is a list of experimental conditions, each of which is represented by a name in square brackets, for example [dark], [light], [density], etc. To build up a model the CORAL software calculates correlation weights of conditions under considering. The sum of the correlation weights is the optimal descriptor for the model one calculated with Eq. (1). Consequently, a potential user can construct own model using solely experimental conditions (without information on molecular structure) together with the CORAL software free available on the Internet. It is clear that it can be not only logP and mutagenicity, but arbitrary endpoints related to nanomaterials. In addition, the approach can be applied to peptides (Toropova and Toropov, 2023a) as well to “simple” organic molecules (Toropov et al., 2023).

3.7. Expectation and reality

It was expected that the new criterion CCCP could improve the predictive potential of the model. In principle, we can say that this is confirmed. At the same time, we have to admit that for anti-correlation the behavior of CCCP is strange. Fortunately, since all the criteria under consideration are aimed at creating correlations, this effect is unlikely to damage models built using the described methodology.

It was expected that the new version of quasi-SMILES would maintain the level of predictive potential of the model. This was confirmed for the cases of IIC and CII. Moreover, by using CCCP, the predictive potential of the model for mutagenicity was noticeably improved.

The application of the Las Vegas algorithm can have several applications. Here a version that involves running the Monte Carlo method multiple times for one distribution into active and passive training sets, a calibration set, and a validation set is considered. However, it is possible to use the Las Vegas algorithm to consider different distributions, with the choice of the best distribution. Another option is to test several versions of the optimal descriptor, choosing the best method. It should be noted that the Monte Carlo method using the CORAL software (<http://www.insilico.eu/coral>) was checked up in simulation of different endpoints for various substances with testing of IIC and CII (Ahmadi et al., 2024; Soleymani et al., 2023; Lotfi et al., 2023; Tajiani et al., 2023; Bhawna et al., 2024). Applying the Las Vegas method and CCCP will probably can make the software more attractive for applications (building up models and databases).

4. Conclusions

The effectiveness of using quasi-SMILES for modeling the behavior of nanomaterials has once again been confirmed. The coefficient of conformism of the correlative prediction (CCCP) is proposed as a new statistical criterion. This criterion allowed us to improve the statistical quality of the models of the octanol/water partition coefficient of gold nanoparticles and the mutagenicity of silver nanoparticles. This ability of the criterion was tested for both endpoints with five random splits of available data into active learning, passive learning, calibration, and

external validation. In addition, the feasibility of using the Las Vegas algorithm for solving practical problems associated with in silico simulation is shown. It is very likely that CCCP is able to improve the predictive potential of other models (other endpoints).

CRediT authorship contribution statement

Alla P. Toropova: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Andrey A. Toropov:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors confirm they have no conflict of interest.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2024.172119>.

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